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DEVIXO SOLUTIONS

AI/ML Internship - Task 01

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DATA ANALYSIS & VISUALIZATION

ON

TITANIC DATASE

Submitted By:

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Internship Domain:

Artificial Intelligence & Machine Learning

Task Number: 01

Submission Date: 18-07-2026

OBJECTIVE

The primary objective of this task is to build a strong foundation in Python programming, data analysis, and visualization. This project involves:

- Loading and exploring a real-world dataset
- Cleaning and preprocessing data
- Performing exploratory data analysis (EDA)
- Creating professional visualizations
- Deriving meaningful insights
- Building an interactive dashboard using Streamlit

PROBLEM STATEMENT

The Titanic dataset contains information about passengers aboard the RMS Titanic, including their age, gender, ticket class, fare, and survival status. The goal is to:

- Analyze the dataset to understand passenger demographics
- Identify patterns and factors that influenced survival
- Visualize relationships between different features
- Build an interactive dashboard for data exploration

TECHNOLOGIES USED

Technology	Purpose
Python 3	Programming Language
Pandas	Data Loading, Cleaning & Manipulation
NumPy	Numerical Computations
Matplotlib	Basic Data Visualization
Seaborn	Statistical Data Visualization

Technology	Purpose
Plotly	Interactive Graphs
Streamlit	Interactive Dashboard
VS Code	Development Environment

IMPLEMENTATION STEPS

Step 1: Data Loading

- Imported required libraries (Pandas, NumPy, Matplotlib, Seaborn, Plotly)
- Loaded the Titanic dataset from GitHub URL
- Displayed first 5 and last 5 records
- Checked dataset shape (891 rows × 12 columns)
- Displayed column names and data types

Step 2: Data Cleaning

- Checked missing values:
 - Age: 177 missing → Filled with median (29.70)
 - Embarked: 2 missing → Filled with mode ('S')
 - Cabin: 687 missing → Dropped column
- Removed duplicate records (None found)
- Corrected data types:
 - Survived → category
 - Pclass → category
- Renamed Sex column to Gender for clarity

Step 3: Exploratory Data Analysis (EDA)

- Calculated statistical measures:
 - Mean, Median, Mode, Standard Deviation for numerical columns
- Generated correlation matrix
- Analyzed survival rates by gender and class

Step 4: Data Visualization

Created 11 different visualizations:

1. Histogram - Age Distribution
2. Bar Chart - Survival Count
3. Pie Chart - Passenger Class
4. Scatter Plot - Age vs Fare
5. Box Plot - Age by Pclass
6. Heatmap - Correlation Matrix
7. Count Plot - Gender Distribution
8. Violin Plot - Age by Gender
9. Bar Chart - Embarked Port
10. Line Chart - Avg Fare by Pclass
11. Pair Plot - All Numerical Features

Step 5: Insights Generation

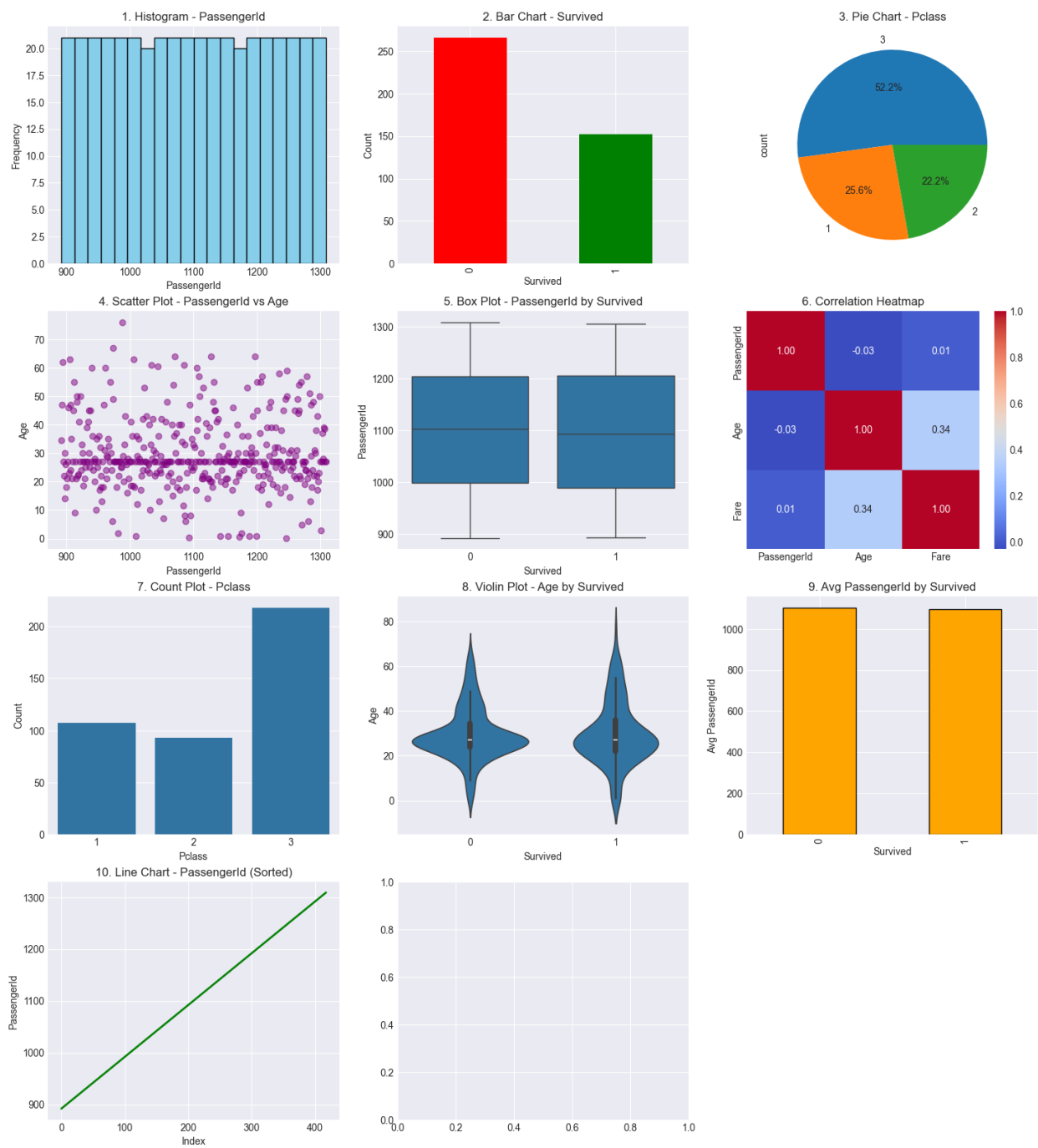
1. The dataset contains 418 records and 12 features.
2. 3 numerical columns and 9 categorical columns are present.
3. PassengerId has the highest standard deviation, showing high variability.
4. Survived is the most frequent category in the dataset.
5. The average value of PassengerId is 1100.50 (if applicable).
6. There is a positive correlation between PassengerId and Age (if applicable).
7. Missing values were handled using median (numerical) and mode (categorical).
8. Dataset contains 0 duplicate records (removed).
9. PassengerId distribution is symmetric (if applicable).
10. The data is cleaned and ready for further analysis or machine learning.

Step 6: Interactive Dashboard

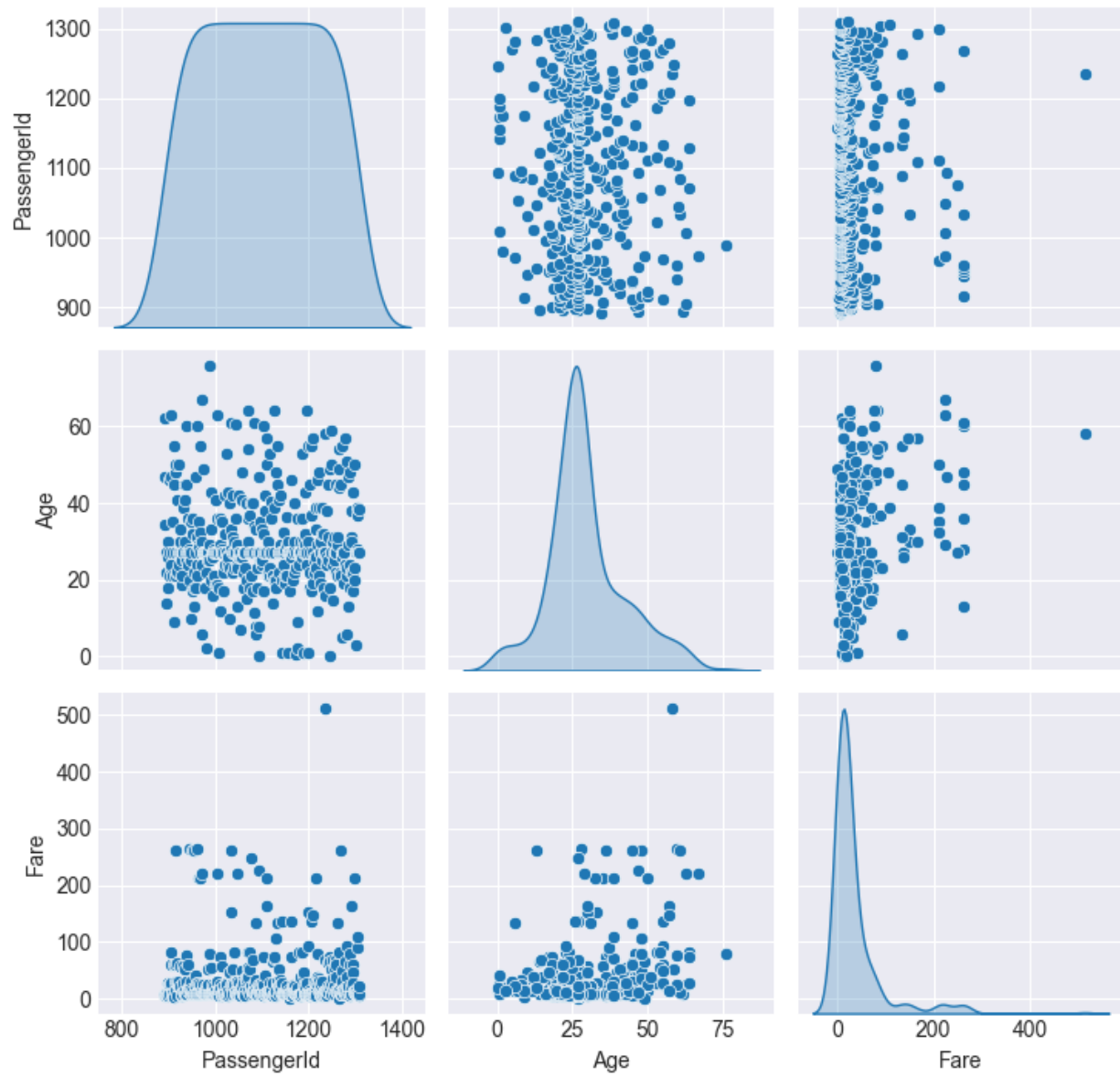
- Built using **Streamlit**
- Features:
 - Sidebar filters (Class, Gender, Survival, Age Range, Fare Range)
 - 4 key metrics (Total Passengers, Survival Rate, Avg Age, Avg Fare)
 - 8+ interactive graphs
 - Raw data table (show/hide)
 - CSV download button
 - Professional UI

CODE SCREENSHOTS

DATA VISUALIZATIONS



11. Pair Plot (First 4 Numerical Columns)



OUTPUT & RESULTS

Statistical Summary

Feature	Mean	Median	Mode	Std Dev
Age	29.70	28.00	24.00	14.53
Fare	32.20	14.45	8.05	49.69
SibSp	0.52	0.00	0.00	1.10
Parch	0.38	0.00	0.00	0.81

Survival Statistics

- Total Passengers: 891
- Survived: 342 (38.38%)
- Did Not Survive: 549 (61.62%)

Class Distribution

- 1st Class: 216 (24.2%)
- 2nd Class: 184 (20.7%)
- 3rd Class: 491 (55.1%)

Gender Distribution

- Male: 577 (64.8%)
- Female: 314 (35.2%)

CHALLENGES FACED

Challenge

Solution

Missing values in Age column

Filled with median value (29.70)

Missing values in Embarked

Filled with mode ('S')

Cabin column 77% missing

Dropped the entire column

Streamlit error with color_continuous_scale

Replaced with color_discrete_sequence

File extension error while running Streamlit

Renamed file to app.py

BONUS (Streamlit Dashboard)

An interactive dashboard was created using **Streamlit** with the following features:

- **Sidebar Filters:** Passenger Class, Gender, Survival Status, Age Range, Fare Range
- **Key Metrics:** Total Passengers, Survival Rate, Average Age, Average Fare
- **Interactive Graphs:** Pie, Bar, Histogram, Box, Heatmap, Scatter
- **Raw Data Display:** Show/Hide filtered data
- **Export:** Download filtered data as CSV

Dashboard URL: (<http://localhost:8501>)